Unit 4

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2.1-2.2

Dimensions of subspaces

Thm Let $W \subseteq V$. If V is finite dimensional, then $\dim V \leq \dim W$

Pf: $\dim V = n$

Let $\beta V = \{u_1, u_2, \dots u_n\}$ be a basis for V.

Let $\mathcal{U} = \{w_1, \dots w_k\} \subset W$ be linearly independent.

using the Replacement Theorem, $S = \beta V$, $U = \mathcal{U} \Rightarrow k \leq n$.

recursively: keep adding linearly independent components until you find the span of W.

Linear Transformations

Def A map T from V to W $(T: V \mapsto W)$ is linear for all $v_1, v_2 \in V$ and $c \in \mathbb{R}$

(a) $T(v_1 + v_2) = T(v_1) + T(v_2)$

(b) $T(cv_1) = cT(v_1)$

Rmk T is linear iff

(c) $T(cv_1 + cv_2) = T(v_1) + cT(v_2)$

Rmk linear combination $T(a_1u_1 + a_2u_2 + \dots + a_ku_k)$

$= a_1T(u_1) + a_2T(u_2) + \dots + a_kT(u_k)$

Linear transforms:

1. $T_0: V \to W$ $u \mapsto \bar{0}_W$

$T_0(v_1 + cv_1) = \bar{0}_W$

$T_0(v_1) + cT_0(v_2) = \bar{0}_W$

2. $T_1: V \to V$ (identity map) $u \mapsto u$

3. $T: \mathbb{R}^2 \to \mathbb{R}^2$ $(x, y) \to (-y, x)$

$T((x_1, y_1) + c(x_2, y_2)) = T(x_1 + cx_2, y_1 + cy_2) = (-y_1 - cy_2, x_1 + cx_2)$

$T(x_1, y_1) + cT(x_2, y_2) = (-y_1, x_1) + (-cy_2, cx_2) = (-y_1 - cy_2, x_1 + cx_2)$

4. $T: P \to P$ $f(x) \mapsto f'(x)$

Pf using property of derivatives

5. $A \in M_{m \times n}$

$L_A: \mathbb{R}^n \to \mathbb{R}^m$ ← if V and W are finite dimensional, any map can be written as a matrix

$\bar{x} \mapsto A\bar{x}$

6. $T_a^b: C^0(\mathbb{R}) \to \mathbb{R}$ ← infinite dimensional ← 1 dimensional

$f \mapsto \int_a^b f(x)\,dx$

Null Space and Range

everything that is mapped to $\bar{0}_W$

Def The null space / kernel of T is $N(T) = \{v \in V \mid T(v) = \bar{0}_W\} \subset V$

Def The range / image of T is $R(T) = \{T(v) \mid = v \in V\} \subset W$

E.g. $N(T_a^b) = \{$ functions in $C^0(\mathbb{R})$ whose definite integral is $0\}$.

$R(T_a^b) = \mathbb{R}$ ← T_a^b is onto

Subspaces:

Thm if T is linear then $N(T)$ is a subspace of V and $R(T)$ is a subspace of W.

Pf: $R(T)$

1. $\bar{0}_W \in R(T) \to v \in V$ so $T(v) = \bar{0}_W$

Since T is linear, $T(\bar{0}_V) = \bar{0}_W$

2. Closed under addition:

Need $w_1, w_2 \in R(T) \Rightarrow w_1 + w_2 \in R(T)$

$w_1 = T(v_1)$, $w_2 = T(v_2)$

$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2)$

$w_1 + w_2 \in R(T)$

3. Closed under multiplication:

Need $w \in R(T) \Rightarrow cw \in R(T)$

$w = T(v)$ $cw = cT(v) = T(cv)$

Dimension Theorem / Rank-Nullity Theorem

Thm If $T: V \to W$ is linear and V is finite dimensional, then

$\dim(\underset{\text{nullity}}{N(T)}) + \dim(\underset{\text{rank}}{R(T)}) = \dim(V)$

Ex $\mathbb{R}^n \to \mathbb{R}^n$

$(x_1, \dots x_n) \mapsto (x_1, \dots, x_m, 0, \dots 0)$ $(m < n)$

$N(T) = \{(x_1, \dots, x_n) \mid x_1 = 0, x_2 = 0, \dots x_m = 0\}$

$\dim(N(T)) = \underline{n - m}$

$R(T) = \{(x_1, \dots, x_m, 0, \dots, 0)\}$

$\dim(R(T)) = m$

$\dim N(T) + \dim(R(T)) = n - m + m = n = \dim(\mathbb{R}^n)$

Pf $\dim V = n$

$N(T)$ is a subspace of V of $\dim k \leq n$

Let $\beta_N = \{u_1, \dots, u_k\}$ be a basis of $N(T)$.

1. Extend β_N by $n-k$ vectors to form a basis of V.

Use replacement and refinement Theorem

2. $\{T(v_1), \dots, T(v_{n-k})\}$ is a basis for $R(T)$

Prove linear independence and span

So $\dim(N(T)) + \dim(R(T)) = k + n - k = n = \dim V$

1-1 linear transformations

Def $T: V \to W$ is onto if $R(T) = W$ or $\forall w \in W, \exists v \in V$ st $T(v) = w$

Def $T: V \to W$ is 1-1 if $T(v_1) = T(v_2) \Rightarrow v_1 = v_2$

Thm If $T: V \to W$ is linear, then T is 1-1 iff $N(T) = \{\bar{0}_V\}$

Thm Let $T: V \to W$ be a linear map between vectors spaces of the same finite dimension. Then the following are equivalent (each imply the other)

(a) T is 1-1

(b) T is onto

(c) $\underset{\dim(R(T))}{\operatorname{rank}(T)} = \dim V$

Matrix representation of linear transforms

Linear map: $A \in M_{m \times n}$

$L_A: \mathbb{R}^n \to \mathbb{R}^m$ $(\bar{x} \to A\bar{x})$

Thm Any linear map between finite dimensional spaces can be represented by a matrix

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Coordinate expression for a vector

We can express every $v \in V$ uniquely as an element of Span(β)

Def $[v]_\beta = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$ ← dim $= n$

Generalise: Each basis β of V ($\dim V = n$) has a map $T: V \to \mathbb{R}^n$.

$T: V \to W$ is linear, V, W are finite dimensional.

Let $\beta = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$

$T(v) = a_1w_1 + \dots + a_mw_m$

For v_j, $T(v_j) = a_{ij}w_1 + a_{2j}w_2 + \dots + a_{mj}w_m$

Def $[T]_\beta^\gamma$ is the matrix representation of T wrt to basis β, γ

$$[T]_\beta^\gamma = \begin{pmatrix} | & & | \\ [T(v_1)]_\gamma & \dots & [T(v_n)]_\gamma \\ | & & \end{pmatrix}$$

Thm For all $v \in V$, $[T(v)]_\gamma = \underset{\text{map}}{[T]_\beta^\gamma} [v]_\beta$

Ex. $T_d: P_3 \to P_2$

$f \to f'$

$\beta = \{1, x, x^2, x^3\}$, $\gamma = \{1, x, x^2\}$

$[T_d(1)]_\beta = [0]_\beta = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ $[T_d(x^2)]_\beta = [2x]_\beta = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

$[T_d(x)]_\beta = [1]_\beta = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $[T_d(x^3)]_\beta = [3x^2]_\beta = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$

$$[T_d]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$